Blessed G/Michael Catholic Secondary School

1st Semester Mathematics Test Two for Grade 10th

Dec, 2018 E.C. Time Allowed: 1:20'

Name: _____ Sec: 10th _____ R.No: _____

I. Choose the best answer among the given alternatives.

____ **1.** The Simplest form of ((-3y)^2) / ((-y)^5) is :-------

A. 9 B. -9 C. -9y^-3 D. 9 / (y^3)

____ **2.** If log_3(y^2) = 2 then what is the value of y ?

A. 9 B. -9 C. 3 D. 0

____ **3.** If 4^x = 1/16 then what is x ?

A. 1 B. -1 C. -2 D. 2

____ **4.** The simplest form of 5th-root(243) is :----------

A. 1/3 B. 3 C. 5 D. 1/5

____ **5.** If (9^x)(4^x) = 6*sqrt(6) then x = ----------.

A. 3 B. 3/4 C. 4/3 D. 4

____ **6.** For any positive real number x, which one is equal to (log_a x)^2 ?

A. 2 log_a x B. log_a (x^2) C. (log_a x) / 2 D. (log_a x) * (log_a x)

____ **7.** For what value of x is the equation 3^(x-1) = sqrt(3) ?

A. 2 B. 1/2 C. 2/3 D. 3/2

____ **8.** If log_a x = log_x a for positive numbers x & a (x, a "` 1) then x = ----------.

A. a^2 B. sqrt(a) C. { } D. 1/a

____ **9.** If 3*(9/25)^(x-1) = 5*(27/125)^x then x = ----------.

A. 0 B. 1 C. 2 D. -1

____ **10.** What is the value of x in log_2 x + log_2(x + 2) = log_2 8 then ?

A. -4 B. 2 C. 6 D. A & B

II. Give short answer only.

11. The simplest form of 3/8 * (log_(1/8) (1/128)) is _____

12. If log_4 x - log_4(x - 1) = log_2 8 then x = _____

13. log_4 x = 1/2 then x = _____

14. What is the simplest form of ((3rd-root(32))^2) / (4^(5/3)) = _____

15. If log 2 = 0.3010 and log 3 = 0.4771 then log_(1/3) 0.0002 = _____

III. Workout (Show all necessary steps clearly)

16. What is the simplest form of log_4 (sqrt(16)) ?

Solution space / working steps:

17. Simplify: ((n^2) * (n^3)) / (n^-2)

Solution space / working steps:

18. If log_2(x - 1) + log_2 3 = 3 then find the value of x.

Solution space / working steps:

19. Evaluate: log 4 + log 25

Solution space / working steps:

20. If log_x 3 = 1/3 then find the value of x.

Solution space / working steps: